**DEPARTMENT OF TELECOMMUNICATION ENGINEERING**
**BACHELOR OF SCIENCE IN CYBER SECURITY**
**MEHRAN UNIVERSITY OF ENGINEERING & TECHNOLOGY, JAMSHORO**


**Network Protocol & Architecture LAB**

**EXPERIMENT # 04**


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Score: ____________ Signature of the Lab Tutor: ________________ Date: ____________

## GVRP Configuration

| LAB PERFORMANCE INDICATOR | SUBJECT KNOWLEDGE | DATA ANALYSIS AND INTERPRETATION | ABILITY TO CONDUCT EXPERIMENT | SCORE |
|---|---|---|---|---|
| OBJECTIVE NO: | | | | |

**Learning Objectives:**

- As a result of this lab section, you should achieve the following tasks:
  - Configuration of GVRP
  - Setting of the GVRP registration mode


**Keywords:** GVRP, VCMP Roles.

**Duration:** 03 hours


## 1 Overview of lab

### 1.1. Introduction

This lab demonstrates the configuration and working of GVRP (GARP VLAN Registration Protocol) for dynamic VLAN propagation across multiple switches using Huawei's eNSP simulator. The objective is to simplify VLAN management by allowing switches to automatically learn and register VLANs without manual configuration on every device, The lab explores VCMP (VLAN Central Management Protocol), a Huawei-specific enhancement for centralized VLAN management. VCMP defines the roles of switches as VCMP Server and VCMP Client, where the server controls VLAN creation and propagation, and clients synchronize VLAN information from the server.

### 1.2. GVRP

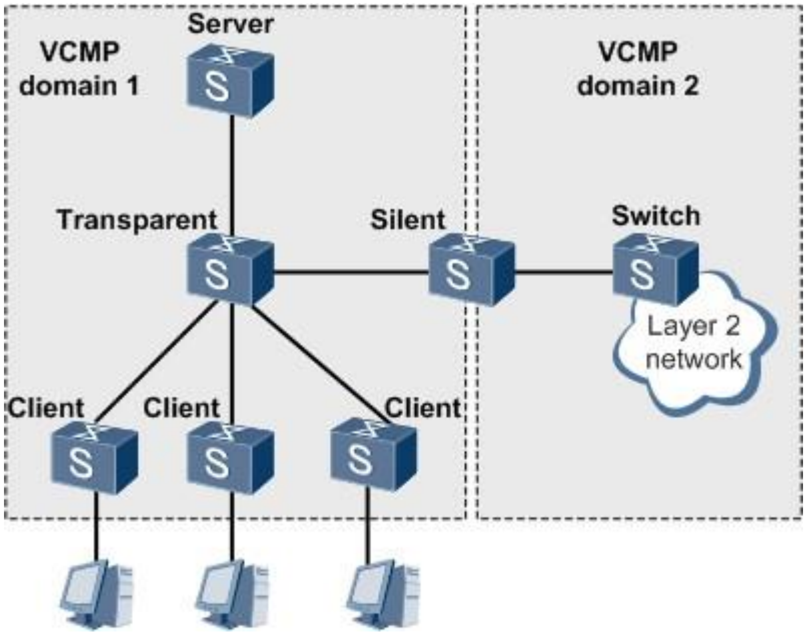
GVRP (GARP VLAN Registration Protocol) is a network rule that helps switches automatically share and learn VLAN information from each other**.** If you create a VLAN on one switch and GVRP is enabled, that switch will automatically tell other connected switches about this VLAN. Those switches can then create the same VLAN without you having to configure them manually.

- To automatically share VLAN information between switches.
- To reduce manual VLAN configuration on every switch.
- To ensure consistency of VLANs across the network.
- To save time in large networks where many VLANs and switches are used.

- To simplify network management, especially in dynamic environments.

### 1.3. VCMP CONCEPT

VCMP uses a VCMP domain to manage switches and determine attributes of switches in the VCMP domain based on roles. VCMP defines four roles: server, client, transparent, and silent. Figure below shows VCMP domains and roles in the VCMP domains.




As shown in figure above, a VCMP domain is composed of switches that have the same VCMP domain name and are connected through trunk or hybrid interfaces. All switches in the VCMP domain must use the same domain name, and each switch can join only one VCMP domain. Switches in different VCMP domains cannot synchronize VLAN information.

A VCMP domain specifies the scope for the administrative switch and managed switches. Switches in a VCMP domain are managed by the administrative switch. There is only one administrative switch and multiple managed switches in a VCMP domain.

### VCMP Roles

VCMP determines attributes of switches based on VCMP roles.

| VCMP Role | Function | Remarks |
|---|---|---|
| **Server** | The VCMP server synchronizes VLAN information to other switches in the local VCMP domain. | VLAN information created and deleted on the VCMP server is broadcast in a VCMP domain. |
| **Client** | A VCMP client belongs to a specified VCMP domain and synchronizes VLAN information with the VCMP server. | VLAN information created and deleted on a VCMP client is not broadcast in a VCMP domain, but is overwritten by VLAN information sent by the VCMP server. |
| **Transparent** | A VCMP transparent switch does not affect other switches in the local VCMP domain and is not affected by VCMP management behaviors such as VLAN creation and deletion. | A VCMP transparent switch transparently forwards VCMP packets to only trunk or hybrid links.<br><br>VLAN information created and deleted on a VCMP transparent switch is not affected by the VCMP server and is not broadcast in a VCMP domain.<br><br>In this way, some switches that do not need to be managed by VCMP can forward VCMP packets. |
| **Silent** | Deployed at the edge of a VCMP domain, a VCMP silent switch does not affect other switches in the local VCMP domain and is not affected by VCMP management behaviors. The VCMP silent switch prevents VCMP packets in a VCMP domain from being transmitted to other VCMP domains. | A VCMP silent switch directly discards received VCMP packets.<br><br>VLAN information created and deleted on a VCMP silent switch is not affected by the VCMP server and is not broadcast in a VCMP domain. |

**Note:** VCMP transparent and silent switches do not belong to any VCMP domain.

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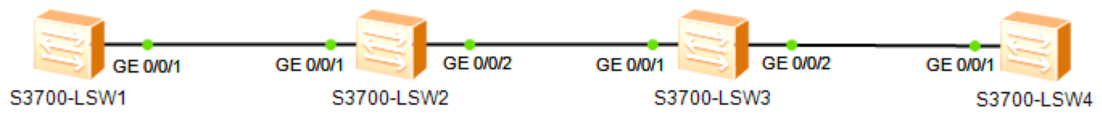
Topology




**Scenario**

The enterprise network contains multiple switches which are expected to be regularly managed. VLANs are required to be applied and removed as necessary on all switches however this tends to be a laborious task for the administrator and often configuration mistakes occur due to human error. The administrator wishes to simplify the VLAN management process and has requested that GVRP be enabled on all switches and the registration mode on the interfaces be set.

## Step 1 Preparing the environment

```
<Huawei> undo terminal monitor
<Huawei> system-view
Enter system view, return user view with Ctrl+Z.
[Huawei] sysname S1
[S1]

<Huawei>
<Huawei> system-view
Enter system view, return user view with Ctrl+Z.
[Huawei] sysname S2
[S2]

<Huawei>
<Huawei> system-view
Enter system view, return user view with Ctrl+Z.
[Huawei] sysname S3
[S3]

<Huawei>
<Huawei> system-view
Enter system view, return user view with Ctrl+Z.
[Huawei] sysname S4
[S4]
```

## Step 2 Configure trunk between the switches

```
[S1]
[S1] interface GigabitEthernet 0/0/1
[S1-GigabitEthernet0/0/1] port link-type trunk
[S1-GigabitEthernet0/0/1] port trunk allow-pass vlan all
[S1-GigabitEthernet0/0/1] q
[S1]

[S2]
[S2] interface GigabitEthernet 0/0/1
[S2-GigabitEthernet0/0/1] port link-type trunk
[S2-GigabitEthernet0/0/1] port trunk allow-pass vlan all
[S2-GigabitEthernet0/0/1] q
[S2] interface GigabitEthernet 0/0/2
[S2-GigabitEthernet0/0/2] port link-type trunk
[S2-GigabitEthernet0/0/2] port trunk allow-pass vlan all
[S2-GigabitEthernet0/0/2] q
[S2]

[S3]
[S3] interface GigabitEthernet 0/0/1
[S3-GigabitEthernet0/0/1] port link-type trunk
[S3-GigabitEthernet0/0/1] port trunk allow-pass vlan all
[S3-GigabitEthernet0/0/1] q
[S3]
[S3] interface GigabitEthernet 0/0/2
[S3-GigabitEthernet0/0/2] port link-type trunk
[S3-GigabitEthernet0/0/2] port trunk allow-pass vlan all
[S3-GigabitEthernet0/0/2] q
[S3]

[S4]
[S4] interface GigabitEthernet 0/0/1
[S4-GigabitEthernet0/0/1] port link-type trunk
[S4-GigabitEthernet0/0/1] port trunk allow-pass vlan all
[S4]
```

## Step 3 Enable GVRP globally, and on all relevant interfaces

```
[S1]
[S1]
[S1] gvrp
[S1] interface GigabitEthernet 0/0/1
[S1-GigabitEthernet0/0/1] gvrp
```

```
[S1-GigabitEthernet0/0/1] q
[S1]

[S2]
[S2] gvrp
[S2] interface GigabitEthernet 0/0/1
[S2-GigabitEthernet0/0/1] gvrp
[S2-GigabitEthernet0/0/1] q
[S2]
[S2] interface GigabitEthernet 0/0/2
[S2-GigabitEthernet0/0/2] gvrp
[S2-GigabitEthernet0/0/2] q
[S2]

[S3]
[S3] gvrp
[S3] interface GigabitEthernet 0/0/1
[S3-GigabitEthernet0/0/1] gvrp
[S3-GigabitEthernet0/0/1] q
[S3]
[S3] interface GigabitEthernet 0/0/2
[S3-GigabitEthernet0/0/2] gvrp
[S3-GigabitEthernet0/0/2] q
[S3]

[S4]
[S4]
[S4] gvrp
[S4] interface GigabitEthernet 0/0/1
[S4-GigabitEthernet0/0/1] gvrp
[S4-GigabitEthernet0/0/1] q
[S4]
```

Run the **display gvrp statistics** command to view GVRP statistics.

```
[S2] display gvrp statistics

  GVRP statistics on port GigabitEthernet0/0/1
    GVRP status                       : Enabled
    GVRP registrations failed         : 0
    GVRP last PDU origin              : 4c1f-cc88-13e7
    GVRP registration type            : Normal

  GVRP statistics on port GigabitEthernet0/0/2
    GVRP status                       : Enabled
```

```
    GVRP registrations failed        : 0
    GVRP last PDU origin             : 4c1f-cc6b-5655
    GVRP registration type           : Normal
[S2]
```

The registration type is set as normal by default. Use the **display vlan** command to view configured VLANs.

```
[S2] display vlan
The total number of vlans is : 1
--------------------------------------------------------------------------------
U: Up;         D: Down;         TG: Tagged;         UT: Untagged;
MP: Vlan-mapping;               ST: Vlan-stacking;
#: ProtocolTransparent-vlan;    *: Management-vlan;
--------------------------------------------------------------------------------
VID  Type    Ports
--------------------------------------------------------------------------------
1     common  UT:Eth0/0/1(D)      Eth0/0/2(D)      Eth0/0/3(D)      Eth0/0/4(D)
Eth0/0/5(D)             Eth0/0/6(D)            Eth0/0/7(D)             Eth0/0/8(D)
Eth0/0/9(D)             Eth0/0/10(D)           Eth0/0/11(D)           Eth0/0/12(D)
Eth0/0/13(D)     Eth0/0/14(D)     Eth0/0/15(D)     Eth0/0/16(D)
                 Eth0/0/17(D)     Eth0/0/18(D)     Eth0/0/19(D)     Eth0/0/20(D)
                 Eth0/0/21(D)     Eth0/0/22(D)     GE0/0/1(U)       GE0/0/2(U)

VID  Status  Property      MAC-LRN Statistics Description
--------------------------------------------------------------------------------
1    enable  default       enable  disable    VLAN 0001
--------------------------------------------------------------------------------
```

Create VLAN 2 and 100 on S1, VLAN 2 on S2 and S3, and VLAN 2 and 200 on S4.

```
[S1]
[S1] vlan batch 2 100
[S1]

[S2]
[S2] vlan 2
[S2-vlan2] q
[S2]

[S3]
[S3] vlan 2
[S3-vlan2] q
[S3]
```

```
[S4]
[S4] vlan batch 2 200
[S4]
```

Use the **display vlan** command to verify the VLAN configuration on S2 and S3.

```
[S2]
[S2] display vlan
The total number of vlans is : 4
--------------------------------------------------------------------------------
U: Up;         D: Down;         TG: Tagged;         UT: Untagged;
MP: Vlan-mapping;               ST: Vlan-stacking;
#: ProtocolTransparent-vlan;    *: Management-vlan;
--------------------------------------------------------------------------------
VID  Type    Ports
--------------------------------------------------------------------------------
1    common  UT:Eth0/0/1(D)     Eth0/0/2(D)     Eth0/0/3(D)     Eth0/0/4(D)
     Eth0/0/5(D)     Eth0/0/6(D)     Eth0/0/7(D)     Eth0/0/8(D)
     Eth0/0/9(D)     Eth0/0/10(D)    Eth0/0/11(D)    Eth0/0/12(D)
     Eth0/0/13(D)    Eth0/0/14(D)    Eth0/0/15(D)    Eth0/0/16(D)
     Eth0/0/17(D)    Eth0/0/18(D)    Eth0/0/19(D)    Eth0/0/20(D)
     Eth0/0/21(D)    Eth0/0/22(D)    GE0/0/1(U)      GE0/0/2(U)

2    common  TG:GE0/0/1(U)      GE0/0/2(U)
100  dynamic TG:GE0/0/1(U)
200  dynamic TG:GE0/0/2(U)


VID  Status  Property      MAC-LRN Statistics Description
--------------------------------------------------------------------------------
1    enable  default       enable  disable    VLAN 0001
2    enable  default       enable  disable    VLAN 0002
     100  enable  default       enable  disable    VLAN 0100
     200  enable  default       enable  disable    VLAN 0200
     ----------------------------------------------------------------------------
     ----

------

[S3]
[S3] display vlan
The total number of vlans is : 4
--------------------------------------------------------------------------------
U: Up;         D: Down;         TG: Tagged;         UT: Untagged;
MP: Vlan-mapping;               ST: Vlan-stacking;
```

```
#: ProtocolTransparent-vlan;     *: Management-vlan;
--------------------------------------------------------------------------------
VID  Type     Ports
--------------------------------------------------------------------------------
1     common   UT:Eth0/0/1(D)      Eth0/0/2(D)      Eth0/0/3(D)      Eth0/0/4(D)
Eth0/0/5(D)      Eth0/0/6(D)      Eth0/0/7(D)      Eth0/0/8(D)
                 Eth0/0/9(D)      Eth0/0/10(D)     Eth0/0/11(D)     Eth0/0/12(D)
Eth0/0/13(D)     Eth0/0/14(D)     Eth0/0/15(D)     Eth0/0/16(D)
Eth0/0/17(D)     Eth0/0/18(D)     Eth0/0/19(D)     Eth0/0/20(D)
Eth0/0/21(D)     Eth0/0/22(D)     GE0/0/1(U)       GE0/0/2(U)

2     common   TG:GE0/0/1(U)       GE0/0/2(U)
100   dynamic TG:GE0/0/1(U)
200   dynamic TG:GE0/0/2(U)
VID  Status   Property      MAC-LRN Statistics Description
--------------------------------------------------------------------------------
1     enable   default       enable   disable    VLAN 0001
2     enable   default       enable   disable    VLAN 0002
100   enable   default       enable   disable    VLAN 0100
200   enable   default       enable   disable    VLAN 0200
--------------------------------------------------------------------------------
```

Switch S2 and S3 are learning VLAN 100 and 200 dynamically, but only in one direction. VLAN 2 has been statistically defined.

**Create VLAN 200 on S1 and VLAN 100 on S4 to enable 2-way propagation.**

```
[S1]
[S1] vlan 200
[S1-vlan200] q
[S1]

[S4]
[S4] vlan 100
[S4-vlan100] q
[S4]
```

Run the **display vlan** command to verify the VLAN configuration.

```
[S2]
[S2] display vlan
The total number of vlans is : 4
--------------------------------------------------------------------------------
U: Up;          D: Down;          TG: Tagged;         UT: Untagged;
MP: Vlan-mapping;                 ST: Vlan-stacking;
```

```
#: ProtocolTransparent-vlan;    *: Management-vlan;
--------------------------------------------------------------------------------
VID  Type    Ports
--------------------------------------------------------------------------------
1    common  UT:Eth0/0/1(D)     Eth0/0/2(D)     Eth0/0/3(D)     Eth0/0/4(D)
                Eth0/0/5(D)     Eth0/0/6(D)     Eth0/0/7(D)     Eth0/0/8(D)
Eth0/0/9(D)           Eth0/0/10(D)          Eth0/0/11(D)          Eth0/0/12(D)
Eth0/0/13(D)          Eth0/0/14(D)          Eth0/0/15(D)          Eth0/0/16(D)
Eth0/0/17(D)          Eth0/0/18(D)          Eth0/0/19(D)          Eth0/0/20(D)
Eth0/0/21(D)    Eth0/0/22(D)    GE0/0/1(U)      GE0/0/2(U)

2    common  TG:GE0/0/1(U)      GE0/0/2(U)
100  dynamic TG:GE0/0/1(U)      GE0/0/2(U)
200  dynamic TG:GE0/0/1(U)      GE0/0/2(U)
VID  Status  Property      MAC-LRN Statistics Description
--------------------------------------------------------------------------------
1    enable  default       enable  disable    VLAN 0001
2    enable  default       enable  disable    VLAN 0002
100  enable  default       enable  disable    VLAN 0100
200  enable  default       enable  disable    VLAN 0200
--------------------------------------------------------------------------------


-------
[S3]
[S3] display vlan
The total number of vlans is : 4
--------------------------------------------------------------------------------
U: Up;         D: Down;         TG: Tagged;         UT: Untagged;
MP: Vlan-mapping;               ST: Vlan-stacking;
#: ProtocolTransparent-vlan;    *: Management-vlan;
--------------------------------------------------------------------------------
VID  Type    Ports
--------------------------------------------------------------------------------
1     common   UT:Eth0/0/1(D)      Eth0/0/2(D)      Eth0/0/3(D)      Eth0/0/4(D)
Eth0/0/5(D)             Eth0/0/6(D)             Eth0/0/7(D)             Eth0/0/8(D)
Eth0/0/9(D)             Eth0/0/10(D)            Eth0/0/11(D)            Eth0/0/12(D)
Eth0/0/13(D)            Eth0/0/14(D)            Eth0/0/15(D)            Eth0/0/16(D)
Eth0/0/17(D)            Eth0/0/18(D)            Eth0/0/19(D)            Eth0/0/20(D)
Eth0/0/21(D)    Eth0/0/22(D)    GE0/0/1(U)      GE0/0/2(U)

2    common  TG:GE0/0/1(U)      GE0/0/2(U)
100  dynamic TG:GE0/0/1(U)      GE0/0/2(U)
200  dynamic TG:GE0/0/1(U)      GE0/0/2(U)
VID  Status  Property      MAC-LRN Statistics Description
```

```
--------------------------------------------------------------------------------
1     enable   default       enable   disable    VLAN 0001
2     enable   default       enable   disable    VLAN 0002
100   enable   default       enable   disable    VLAN 0100
200   enable   default       enable   disable    VLAN 0200
--------------------------------------------------------------------------------
```

The highlighted entries indicate the interfaces that have been added to VLAN 100 and 200 on S2 and S3.

## Step 4 Change the registration type for the interfaces

Change the registration type of Gigabit Ethernet 0/0/2 on S2 to FIXED. The same steps can be performed on Gigabit Ethernet 0/0/1 of S3.

```
[S2]
[S2] interface GigabitEthernet 0/0/2
[S2-GigabitEthernet0/0/2] gvrp registration fixed
```

Run the **display gvrp statistics** command to view the changes.

```
[S2-GigabitEthernet0/0/2] display gvrp statistics interface GigabitEthernet 0/0/2
  GVRP statistics on port GigabitEthernet0/0/2
    GVRP status                        : Enabled
    GVRP registrations failed          : 15
    GVRP last PDU origin               : 4c1f-cc6b-5655
    GVRP registration type             : Fixed
[S2-GigabitEthernet0/0/2]q
[S2]

-------
[S3]
[S3] interface GigabitEthernet 0/0/1
[S3-GigabitEthernet0/0/1] gvrp registration fixed
[S3-GigabitEthernet0/0/1] q
[S3]
```

Run the **display vlan** command to view the effect of the fixed registration type.

```
[S2]
[S2] display vlan
The total number of vlans is : 4
--------------------------------------------------------------------------------
U: Up;         D: Down;         TG: Tagged;         UT: Untagged;
MP: Vlan-mapping;               ST: Vlan-stacking;
```

```
#: ProtocolTransparent-vlan;    *: Management-vlan;
--------------------------------------------------------------------------------
VID  Type     Ports
--------------------------------------------------------------------------------
1     common   UT:Eth0/0/1(D)      Eth0/0/2(D)      Eth0/0/3(D)      Eth0/0/4(D)
Eth0/0/5(D)            Eth0/0/6(D)            Eth0/0/7(D)            Eth0/0/8(D)
Eth0/0/9(D)            Eth0/0/10(D)           Eth0/0/11(D)           Eth0/0/12(D)
Eth0/0/13(D)           Eth0/0/14(D)           Eth0/0/15(D)           Eth0/0/16(D)
Eth0/0/17(D)           Eth0/0/18(D)           Eth0/0/19(D)           Eth0/0/20(D)
Eth0/0/21(D)    Eth0/0/22(D)    GE0/0/1(U)      GE0/0/2(U)

2    common   TG:GE0/0/1(U)       GE0/0/2(U)
100  dynamic TG:GE0/0/1(U)
200  dynamic TG:GE0/0/1(U)
VID  Status   Property      MAC-LRN Statistics Description
--------------------------------------------------------------------------------
1    enable   default       enable  disable    VLAN 0001
2    enable   default       enable  disable    VLAN 0002
100  enable   default       enable  disable    VLAN 0100
200  enable   default       enable  disable    VLAN 0200
--------------------------------------------------------------------------------
```

Configure interface Gigabit Ethernet 0/0/2 on S2 to use the FORBIDDEN registration type. The same steps can be performed on Gigabit Ethernet 0/0/1 of S3.

```
[S2]
[S2] interface GigabitEthernet 0/0/2
[S2-GigabitEthernet0/0/2] gvrp registration forbidden
[S2-GigabitEthernet0/0/2] q
[S2]

[S3]
[S3] interface GigabitEthernet 0/0/1
[S3-GigabitEthernet0/0/1] gvrp registration forbidden
[S3-GigabitEthernet0/0/1] q
[S3]
```

Run the **display gvrp statistics** command to view the changes to GVRP.

```
[S2] display gvrp statistics interface GigabitEthernet 0/0/2

  GVRP statistics on port GigabitEthernet0/0/2
    GVRP status                        : Enabled
    GVRP registrations failed          : 485
    GVRP last PDU origin               : 4c1f-cc6b-5655
```

```
    GVRP registration type          : Forbidden
[S2]
```

The GVRP registration type is set to forbidden on the Gigabit Ethernet 0/0/2 interface.

Run the **display vlan** command to view the effect of the forbidden registration.

```
[S2]
[S2] display vlan
The total number of vlans is : 4
--------------------------------------------------------------------------------
U: Up;          D: Down;          TG: Tagged;          UT: Untagged;
MP: Vlan-mapping;                 ST: Vlan-stacking;
#: ProtocolTransparent-vlan;     *: Management-vlan;
--------------------------------------------------------------------------------
VID  Type    Ports
--------------------------------------------------------------------------------
1      common   UT:Eth0/0/1(D)      Eth0/0/2(D)      Eth0/0/3(D)      Eth0/0/4(D)
Eth0/0/5(D)            Eth0/0/6(D)            Eth0/0/7(D)            Eth0/0/8(D)
Eth0/0/9(D)            Eth0/0/10(D)           Eth0/0/11(D)           Eth0/0/12(D)
Eth0/0/13(D)           Eth0/0/14(D)           Eth0/0/15(D)           Eth0/0/16(D)
Eth0/0/17(D)           Eth0/0/18(D)           Eth0/0/19(D)           Eth0/0/20(D)
Eth0/0/21(D)    Eth0/0/22(D)    GE0/0/1(U)      GE0/0/2(U)

2     common   TG:GE0/0/1(U)
100   dynamic  TG:GE0/0/1(U)
200   dynamic  TG:GE0/0/1(U)
VID   Status   Property      MAC-LRN Statistics Description
--------------------------------------------------------------------------------
1     enable   default       enable   disable    VLAN 0001
2     enable   default       enable   disable    VLAN 0002
100   enable   default       enable   disable    VLAN 0100
200   enable   default       enable   disable    VLAN 0200
--------------------------------------------------------------------------------
```

## Final Configuration

```
<S1> display current-configuration
# sysname
S1 #
vcmp role silent
#
vlan batch 2 100 200
```

```
# gvrp
#
interface GigabitEthernet0/0/1 port
link-type trunk
port trunk allow-pass vlan 2 to 4094
gvrp # return
```

---

```
<S2> display current-configuration
# sysname S2
# vlan batch
2
# gvrp
#
interface GigabitEthernet0/0/1 port
link-type trunk
port trunk allow-pass vlan 2 to 4094
gvrp #
interface GigabitEthernet0/0/2 port
link-type trunk
port trunk allow-pass vlan 2 to 4094 gvrp
gvrp registration forbidden
#
return
```

---

```
<S3> display current-configuration
# sysname S3
# vlan batch
2
# gvrp
#
interface GigabitEthernet0/0/1 port
link-type trunk
port trunk allow-pass vlan 2 to 4094 gvrp
gvrp registration forbidden
#
interface GigabitEthernet0/0/2 port
link-type trunk
port trunk allow-pass vlan 2 to 4094
gvrp # return
```

---

```
<S4> display current-configuration
```

```
# sysname
S4
# #
vcmp role silent
#
vlan batch 2 100 200
# gvrp
#
interface GigabitEthernet0/0/1 port
link-type trunk
port trunk allow-pass vlan 2 to 4094
gvrp # return
```

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## Tasks

Try a different topology and demonstrate your learning.

Upload the `.topo .efz .cfg` files